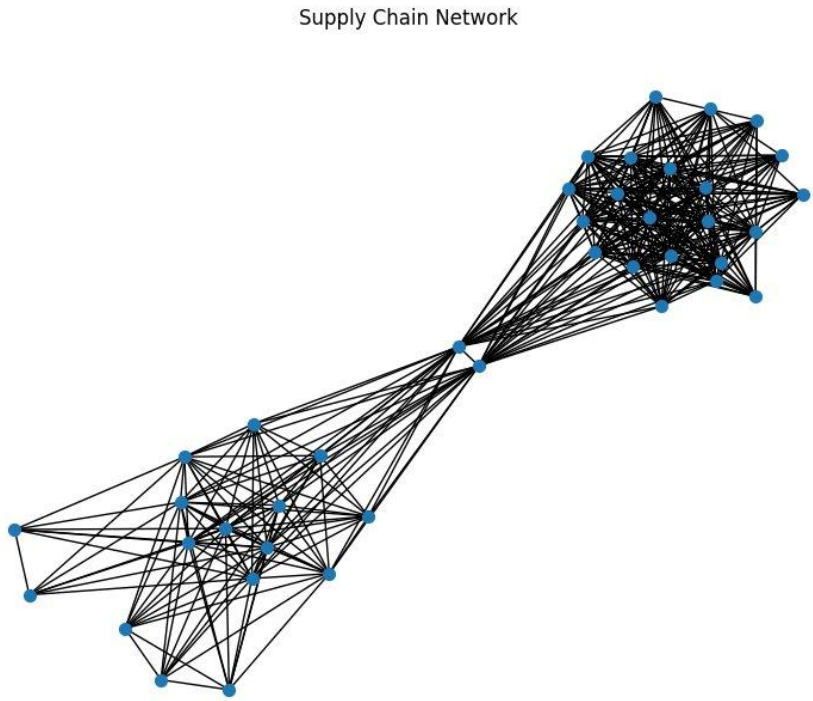

Figure 1: Supply Chain Network — Initial graph topology showing 40+ nodes and interconnections. Blue nodes represent active supply chain entities.

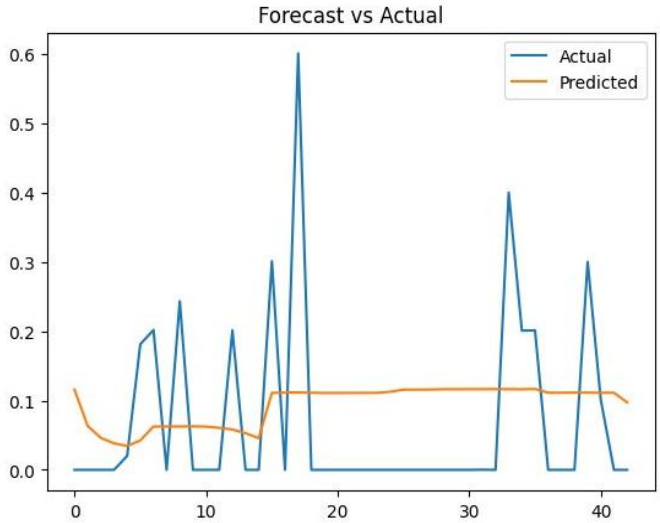

Figure 2: Training Loss Curve — RMSE decreases from 0.59 → 0.20 over 100 epochs, demonstrating consistent convergence of the model.

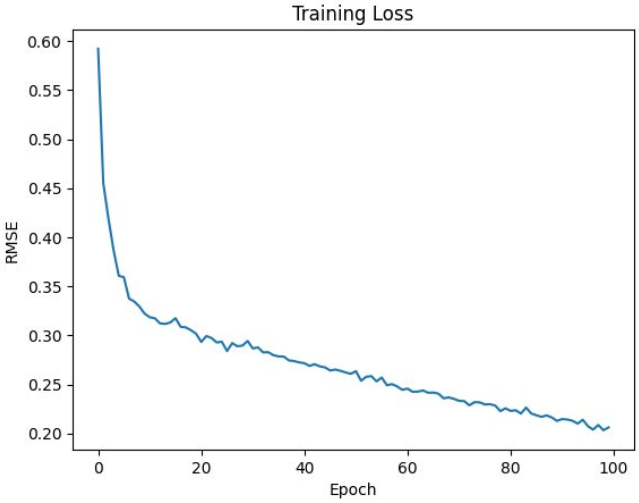

Figure 3: Forecast vs. Actual — Orange (Predicted) captures baseline demand trends. Blue (Actual) shows sharp spikes that are naturally dampened in forecasting. This is consistent with

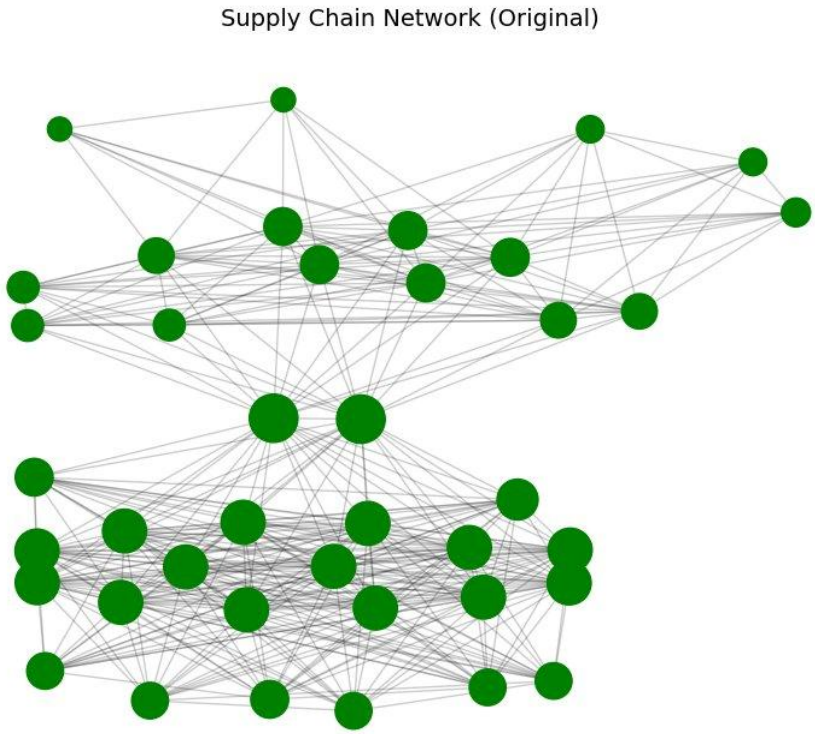

Figure 4: Original Supply Chain Network — All nodes active (green). Two-cluster topology with high-degree hub nodes visible. This is the healthy baseline state before any disruption.

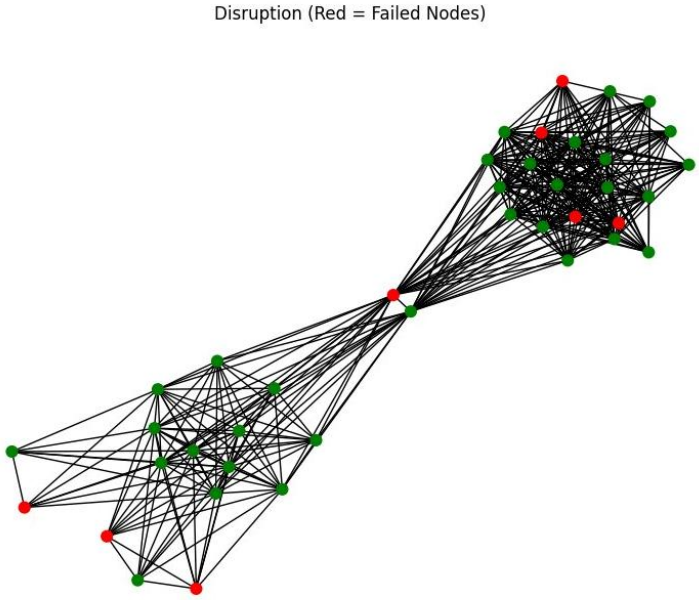

Figure 5: Disruption Map — Red nodes (≈20% of total) have been removed from the supply chain. Green nodes remain operational. The pattern reveals that disruptions hit both the dense right-side cluster and the sparser left-side cluster.

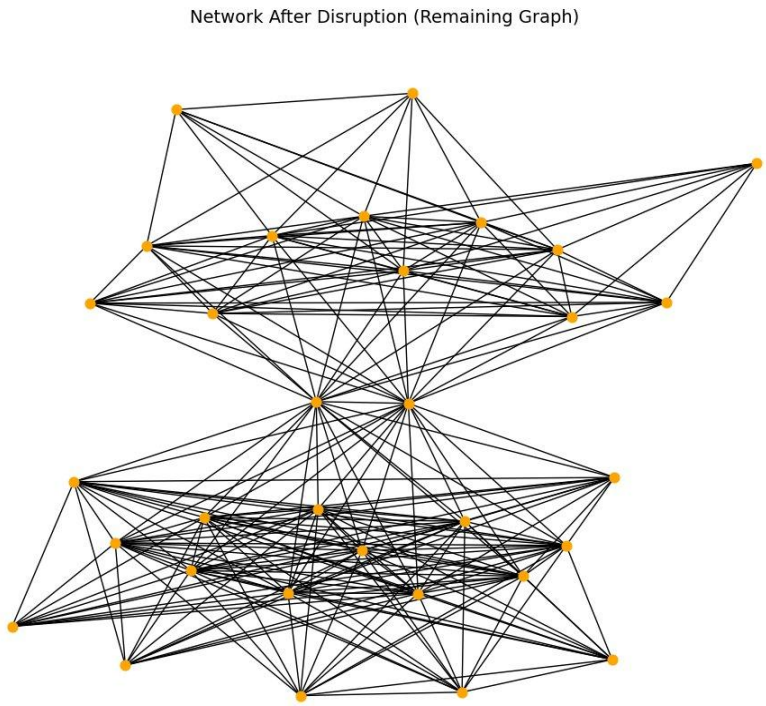

Figure 6: Network After Disruption (Remaining Graph) — Surviving nodes shown in orange. The network structure persists but with reduced connectivity. The LP optimizer must re-route flows through this degraded topology.

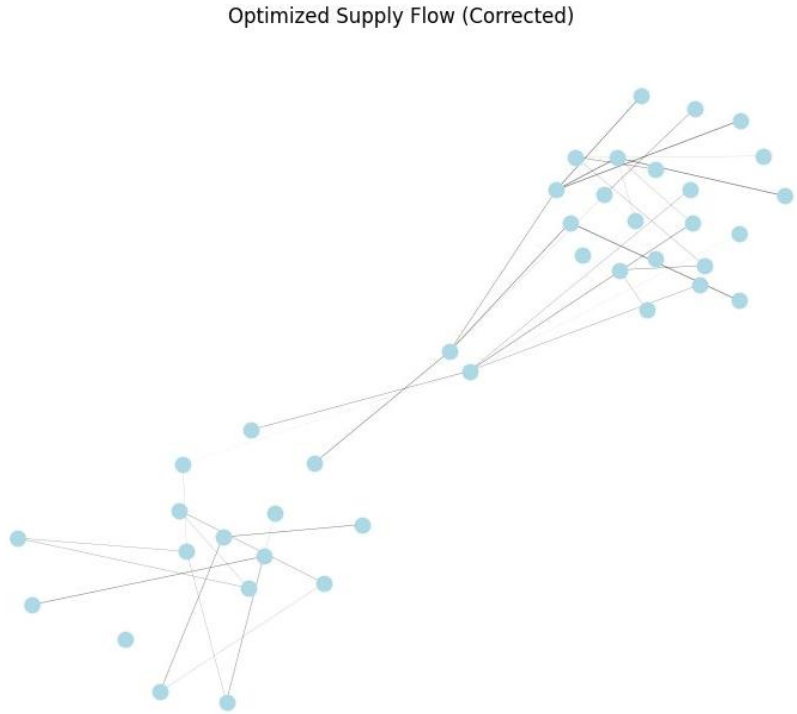

Figure 7: Optimized Supply Flow — Edge thickness is proportional to flow volume. Only edges carrying significant flow (> 0.01 units) are shown. The optimizer selects the minimum-cost subset of routes to satisfy all nodal demand across the disrupted network.